

Course Title: Microcomputer Application I
Course Code: CSE-2402

Topic: Microsoft Excel topics
(Functions, Cross-Sheet Calculations)

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Today's Agenda

Two core topics — Functions and working with data across multiple sheets

01

What is a Function?

Syntax, arguments, categories overview

02

Math & Statistical Functions

SUM, AVERAGE, COUNT, COUNTIF, SUMIF, MAX, MIN...

03

Text & Logical Functions

IF, AND, OR, Nested IF, CONCAT, LEN, LEFT, RIGHT

04

Lookup Functions

VLOOKUP, HLOOKUP, INDEX, MATCH — finding data fast

05

Different Sheet References

Syntax for referencing cells across worksheets

06

Cross-Sheet Calculations

SUM across sheets, 3-D references, worked examples

07

Practice Questions

6 questions testing both topics

08

Lab Task

Multi-sheet Sales Report — 45 minutes

What is an Excel Function?

Topic 01 — Functions are pre-built formulas that perform specific calculations

Definition:

A function is a pre-written formula built into Excel that takes inputs (called arguments), performs a calculation, and returns a result. Functions save time and reduce errors compared to writing complex formulas manually.

Function Anatomy — Colour Coded:

```
=SUMIF( A2:A20 , "Laptops" , B2:B20 )
```

= s: Function
(require name)

Arg 1: range
to check

Arg 2: criteria
(condition)

Arg 3: sum
range



Always starts with =



Arguments go inside ()



Multiple args separated by
,



Some args are optional []

Function Categories Diagram

Excel has 400+ functions — organised into these main categories



Math & Trig

SUM

SUMIF

ROUND

MOD

ABS

POWER

SQRT



Statistical

AVERAGE

MAX

MIN

COUNT

COUNTIF

LARGE

MEDIAN



Text

CONCAT

LEFT

RIGHT

MID

LEN

UPPER

TRIM



Lookup & Ref

VLOOKUP

HLOOKUP

INDEX

MATCH

INDIRECT



Logical

IF

AND

OR

NOT

IFS

IFERROR



Date & Time

TODAY

NOW

DATE

YEAR

MONTH

DATEDIF

Math & Statistical Functions

Topic 02 — The most frequently used functions in Excel

Sample Data

	Name	Dept	Score
2	Ali	Math	85
3	Sara	Science	72
4	Omar	Math	91
5	Nadia	English	60
6	Khalid	Science	78
7	Layla	Math	88
8	Reem	English	55

Data range B2:B9 / D2:D9

SUM

=SUM(D2:D9)

Total of all scores → 622

AVERAGE

=AVERAGE(D2:D9)

Mean score of all students → 77.75

MAX

=MAX(D2:D9)

Highest score in the class → 93

MIN

=MIN(D2:D9)

Lowest score in the class → 55

COUNT

=COUNT(D2:D9)

Number of students with scores → 8

COUNTIF

=COUNTIF(C2:C9,"Math")

Count students in Math dept → 3

SUMIF

=SUMIF(C2:C9,"Math",D2:D9)

Total score for Math dept only → 264

LARGE

=LARGE(D2:D9,2)

2nd highest score → 91



AVERAGEIF syntax: =AVERAGEIF(range, criteria, average_range) — averages only cells matching a condition, e.g. =AVERAGEIF(C2:C9,"Math",D2:D9) → 88

COUNTIF & SUMIF — Conditional Functions

Count or sum only the cells that meet your criteria

COUNTIF — Count with a Condition

Syntax:

```
=COUNTIF( range , criteria )
```

Examples:

```
=COUNTIF(C2:C9, "Math")
```

→ Counts students in Math dept

```
=COUNTIF(D2:D9, ">80")
```

→ Counts scores greater than 80

```
=COUNTIF(D2:D9, ">=60")
```

→ Counts scores 60 or above (Pass)

```
=COUNTIF(B2:B9, "Ali")
```

→ Counts how many times Ali appears

SUMIF — Sum with a Condition

Syntax:

```
=SUMIF( range , criteria , sum_range )
```

Examples:

```
=SUMIF(C2:C9, "Math", D2:D9)
```

→ Total scores for Math dept

```
=SUMIF(D2:D9, ">80", D2:D9)
```

→ Sum all scores above 80

```
=SUMIF(C2:C9, "Science", D2:D9)
```

→ Sum Science dept scores

```
=SUMIF(D2:D9, "<60", D2:D9)
```

→ Sum all failing scores

Logical Functions — IF, AND, OR & Nested IF

Topic 03 — Make decisions with your data

IF Function Syntax:

```
=IF( logical_test , value_if_true , value_if_false )
```

Simple IF

```
=IF(D2>=50,"Pass","Fail")
```

If score in D2 is 50 or above → shows Pass,
otherwise → shows Fail

Score	Formula Result
85	Pass
40	Fail
60	Pass

IF + AND

```
=IF(AND(D2>=50,D2<=100),"Valid","Check")
```

Both conditions must be true: score must be
≥50 AND ≤100

Score	Result
85	Valid
40	Check
110	Check

IF + OR

```
=IF(OR(D2<40,D2>95),"Flag","OK")
```

Either condition true: flag if below 40 OR above
95 (outlier check)

Score	Result
35	Flag
97	Flag
75	OK

Nested IF

```
=IF(D2>=90,"A",IF(D2>=75,"B",IF(D2>=60,"C","D")))
```

Checks multiple thresholds: A for 90+, B for
75+, C for 60+, else D

Score	Grade
93	A
78	B
62	C
45	D

Text Functions

Clean, extract and transform text in your spreadsheet

CONCAT

`=CONCAT(A2," ",B2)`

Join two or more text values into one cell

IN: "Ali" "Hassan"

OUT: "Ali Hassan"

RIGHT

`=RIGHT(A2,4)`

Extract last N characters from the right

IN: "Computer"

OUT: "uter"

LEN

`=LEN(A2)`

Count the number of characters in a cell

IN: "Computer"

OUT: 8

LOWER

`=LOWER(A2)`

Convert all text to lowercase

IN: "HELLO"

OUT: "hello"

LEFT

`=LEFT(A2,3)`

Extract first N characters from the left

IN: "Computer"

OUT: "Com"

MID

`=MID(A2,3,4)`

Extract N characters starting from position

IN: "Computer"

OUT: "mpu"

UPPER

`=UPPER(A2)`

Convert all text to UPPERCASE

IN: "hello"

OUT: "HELLO"

TRIM

`=TRIM(A2)`

Remove extra spaces from start and end

IN: " hello "

OUT: "hello"

VLOOKUP & HLOOKUP

Topic 04 — Find and retrieve data from a table automatically

For VLOOKUP lookup value must present on the 1st column and for HLOOKUP lookup value must present on the 1st row.

VLOOKUP — Vertical Lookup

Syntax:

```
=VLOOKUP(lookup_val, table, col_index, 0)
```

lookup_val The value you are searching for (e.g. Student ID)

table The entire table range that contains your data

col_index Which column number to return (1=first, 2=second...)

0 0 = Exact match 1 = Approximate match (sorted data)

Example:

```
=VLOOKUP(G2, A2:D20, 3, 0)
```

Looks up the value in G2 inside column A of the table A2:D20, and returns the value from column 3 (exact match)

HLOOKUP — Horizontal Lookup

Syntax:

```
=HLOOKUP(lookup_val, table, row_index, 0)
```

Key Difference:

VLOOKUP Searches DOWN a column Returns value from right

HLOOKUP Searches ACROSS a row Returns value from below

When to use which:

- Use VLOOKUP when your data is arranged in columns (most common)
- Use HLOOKUP when your data is arranged in rows (e.g. a horizontal schedule)
- Use INDEX+MATCH for more flexibility — works in both directions

Working with Different Sheets

Topic 05 — Referencing data across multiple worksheets

What & Why:

A workbook can contain multiple worksheets (tabs). You can write formulas that pull data from one sheet into another, or perform calculations that combine data from many sheets — without copying data.

Book1.xlsx

	Product	Units	Revenue
1	Laptop	12	\$14,400
2	Mouse	85	\$2,125
3	Keyboard	30	\$2,250

Data in January sheet

Referenced in Summary sheet

January

February

March

Summary

+



Monthly → Summary



Branch → Total



Dept → HR Report



Product → Invoice

Cross-Sheet Reference Syntax

How to write a formula that reads from a different worksheet

The Syntax Rule:

=SheetName!CellAddress e.g. **=January!D5**

Sheet name (yellow)	Exclamation mark = separator	Cell address (normal reference)
------------------------	------------------------------------	------------------------------------

Sheet name has spaces?

Wrap the sheet name in single quotes: ='Sales Data'!B3 or =SUM('Q1 Report'!D2:D20)

Single cell

=January!D5

Pull the value of cell D5
from the January sheet

Range in formula

=SUM(January!D2:D20)

Sum all values in D2:D20
on the January sheet

Range with spaces

=SUM('Sales Data'!B2:B50)

Sum a range from a sheet
whose name has a space

Across 3 sheets

=SUM(Jan:Mar!D5)

3-D reference: sum cell D5
across all sheets Jan to
Mar

Cell + formula

=January!D5*1.15

Multiply a cross-sheet
value by 1.15 (add 15%
tax)

AVERAGE cross-sheet

=AVERAGE(January!D2:D20)

Average of values from
another sheet

Worked Example — Multi-Sheet Sales Summary

Three monthly sheets + one Summary sheet pulling everything together

January	
Product	Sales
Laptop	14400
Mouse	2125
Keyboard	2250
Total	18775

February	
Product	Sales
Laptop	11200
Mouse	1800
Keyboard	1900
Total	14900

March	
Product	Sales
Laptop	16600
Mouse	2400
Keyboard	2650
Total	21650

Summary Sheet — formulas referencing all three monthly sheets				
Month	Laptop	Mouse	Keyboard	Total Formula
January	14,400	2,125	2,250	=SUM(January!B2:B4)
February	11,200	1,800	1,900	=SUM(February!B2:B4)
March	16,600	2,400	2,650	=SUM(March!B2:B4)

 The formula =SUM(January!B2:B4) in the Summary sheet adds cells B2 to B4 from the January sheet — no copying needed!

3-D References & Creating Cross-Sheet Links

How to write them and how to click-and-link sheets in Excel

3-D References

A 3-D reference spans the same cell or range across multiple consecutive sheets using a colon between sheet names.

```
=SUM(Jan:Dec!B5)
```

Adds cell B5 from every sheet between Jan and Dec (including both)

```
=AVERAGE(Jan:Mar!D2:D20)
```

→ Average of a range across Jan, Feb and Mar sheets

```
=SUM(Sheet1:Sheet5!C3)
```

How to Create by Clicking

1

Click the destination cell

Click the cell where you want the cross-sheet result to appear (e.g. Summary!B2)

2

Type = then your function

Type =SUM(but do NOT press Enter yet

3

Click the source sheet tab

Click the January tab at the bottom of the screen

4

Select the cells you want

Drag to select the range (e.g. B2:B20) on the January sheet

5

Press Enter to confirm

Excel writes the full reference automatically:
=SUM(January!B2:B20)

Common Mistakes with Cross-Sheet References

X =January!D5 (sheet renamed later)

✓ If you rename a sheet, formulas update automatically — but only if the sheet existed first

X Forgetting the ! separator

✓ Without ! Excel cannot tell where the sheet name ends and cell begins

X Forgetting quotes for spaces

✓ ='My Sheet'!A1 — single quotes required whenever the sheet name has a space

X Deleting a referenced sheet

✓ Breaks all formulas referencing it — shows #REF! error

Practice Questions

Test your understanding of Functions and Cross-Sheet Calculations

Q1

MCQ

Which formula correctly counts how many students in C2:C20 belong to the 'Science' department?

- A) =COUNT(C2:C20,"Science") B) =COUNTIF(C2:C20,"Science")
C) =SUM(C2:C20,"Science") D) =IF(C2:C20,"Science")

Q2

Formula

A table has Department names in column A (A2:A50) and Salary in column B (B2:B50). Write a formula that calculates the total salary paid to employees in the 'IT' department only.

Q3

Short Answer

Explain the difference between VLOOKUP and HLOOKUP. In what type of situation would you use each one? Give one real-world example for each.

Q4

Formula

You have three sheets: Term1, Term2, Term3. Cell D10 on each sheet holds the total marks for that term. Write a formula in a Summary sheet that calculates the average across all three terms.

Q5

Short Answer

A student writes the formula =SUM(JanuaryD2:D20) and gets an error. Identify the mistake in this formula and write the corrected version.

Q6

Nested IF

Write a Nested IF formula for cell E2 that assigns grades based on a score in D2:
90 and above → "Distinction" 75–89 → "Merit" 60–74 → "Pass" below 60 → "Fail"

Lab Task — Multi-Sheet Sales Report

Duration: 40–45 minutes | Individual | Submit as Firstname_Lastname_Excel_Lab2.xlsx

Scenario:

You are a sales analyst at a retail company. Build a quarterly sales workbook with three monthly data sheets (January, February, March) and one Summary sheet that uses cross-sheet formulas and functions to produce a full sales report.



Sheet Setup

- ◆ Create 4 sheets: January, February, March, Summary
- ◆ Jan/Feb/Mar: columns Product (A), Units Sold (B), Unit Price (C), Revenue (D)
- ◆ Add 6 products per sheet (Laptop, Phone, Tablet, Mouse, Keyboard, Headset)
- ◆ Fill in Units Sold and Unit Price values for each sheet
- ◆ Column D (Revenue) = Units Sold × Unit Price using formula $=B2*C2$
- ◆ Summary sheet: rows for each month + Quarterly Total row

Functions Required

- ◆ Revenue column: $=B2*C2$ copied down for all products
- ◆ Row 9 (Total per sheet): $=SUM(D2:D7)$ — total monthly revenue
- ◆ Summary B2: $=January!D9$ — pull Jan total
- ◆ Summary B3: $=February!D9$ — pull Feb total
- ◆ Summary B4: $=March!D9$ — pull Mar total
- ◆ Summary B5: $=SUM(January:March!D9)$ — 3-D total
- ◆ Summary: AVERAGE, MAX, MIN of the three month totals



Formatting Required

- ◆ Each sheet: header row green, bold, white text
- ◆ Revenue column: Currency format (\$1,234.00)
- ◆ Units Sold column: Number format, 0 decimal places
- ◆ Total row: bold, dark green fill on all 3 monthly sheets
- ◆ Summary sheet: colour-code each month row (green/teal/sky)
- ◆ Grand Total row: bold, amber fill, larger font (14pt)
- ◆ All sheets: borders on all cells, auto-fit column widths

Marking: Sheet Setup 20 pts + Functions & Formulas 50 pts + Formatting 30 pts = 100 pts

per month

Thank You

Any Questions?